COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/31/2026	**DATE SUBMITTED:**	07/31/2026
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ACTIVITY NO. 4.1 STACKS

1. PROCEDURE OUTPUT

What procedures did you perform?

B.2. Stacks using Linked Lists

```
#include <iostream>

class Node {
public:
    int data;
    Node* next;
};

Node *head = NULL, *tail = NULL;

void push(int newData) {
    Node *newNode = new Node;

    newNode->data = newData;
    newNode->next = head;

    if (head == NULL) {
        head = tail = newNode;
    } else {
        newNode->next = head;
        head = newNode;
    }
}

int pop() {
    int tempVal;
    Node *temp;

    if (head == NULL) {
        head = tail = NULL;
        std::cout << "Stack Underflow." << std::endl;
        return -1;
    } else {
        temp = head;
        tempVal = temp->data;
        head = head->next;
        delete temp;
        return tempVal;
    }
}

void Top() {
    if (head == NULL) {
        std::cout << "Stack is Empty." << std::endl;
        return;
    } else {
        std::cout << "Top of Stack: " << head->data << std::endl;
    }
}

int main() {
    push(1);
    std::cout << "After the first PUSH top of stack is: ";
    Top();

    push(5);
    std::cout << "After the second PUSH top of stack is: ";
    Top();

    pop();
    std::cout << "After the first POP operation, top of stack is: ";
    Top();

    pop();
    std::cout << "After the second POP operation, top of stack is: ";
    Top();

    pop();

    return 0;
}
```

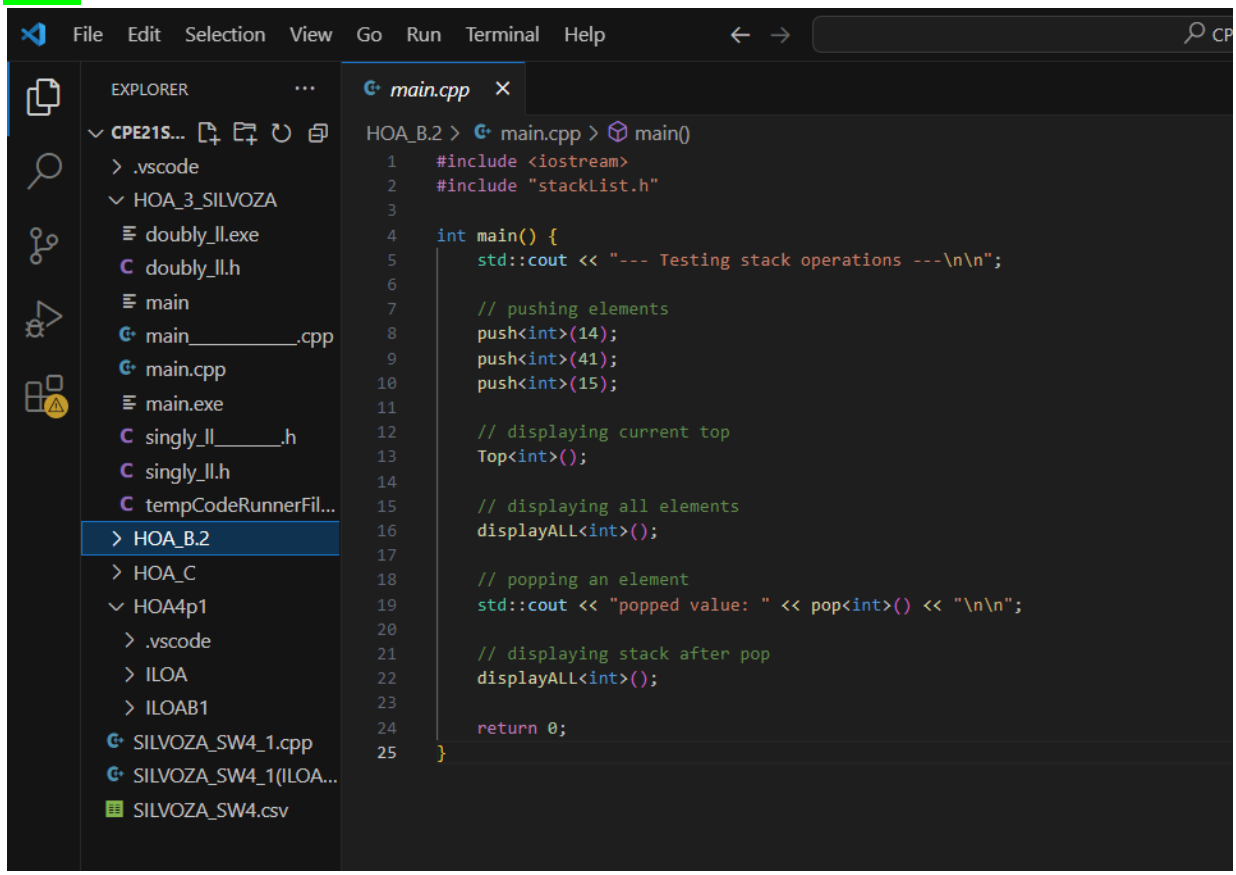
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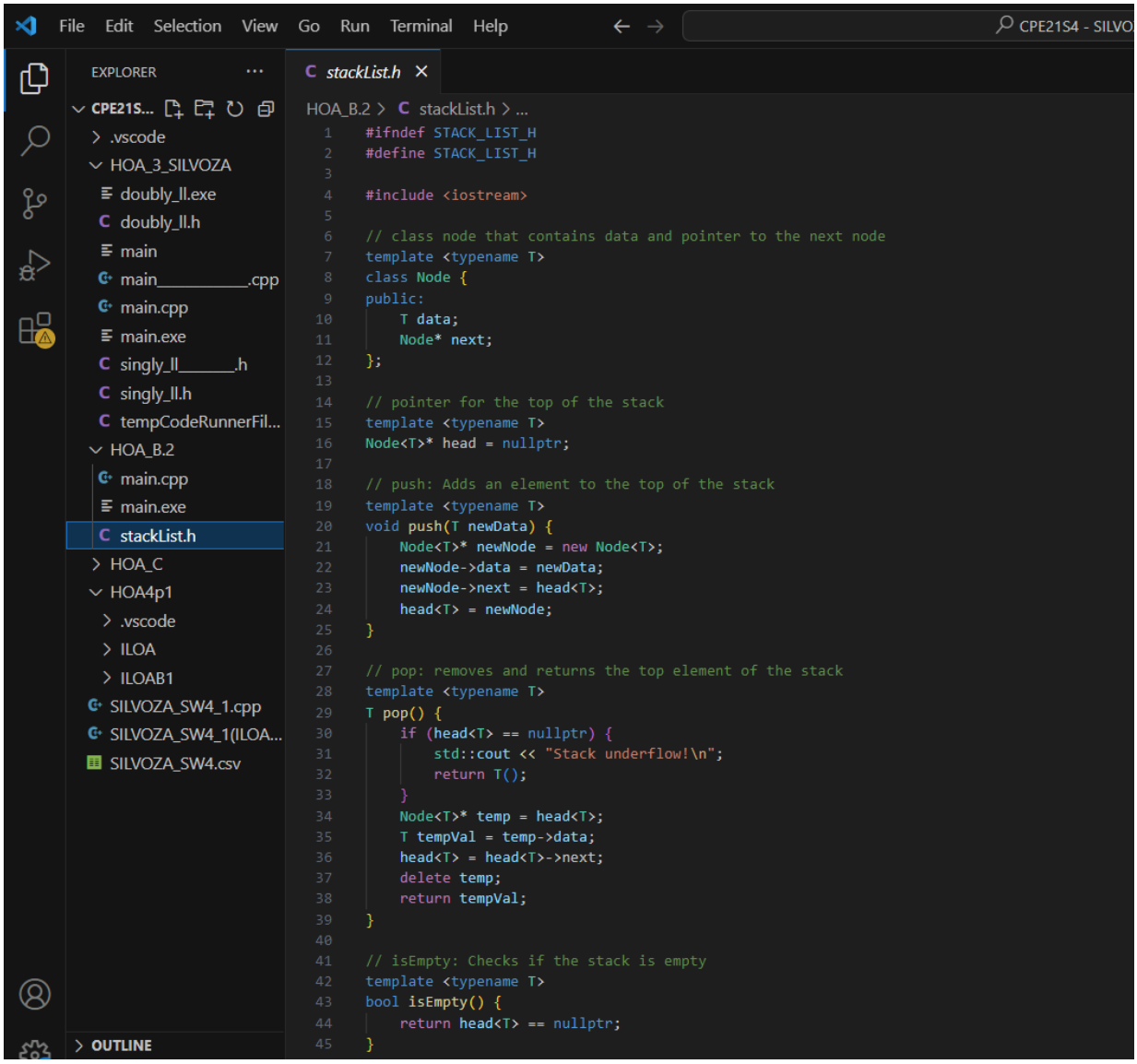
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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

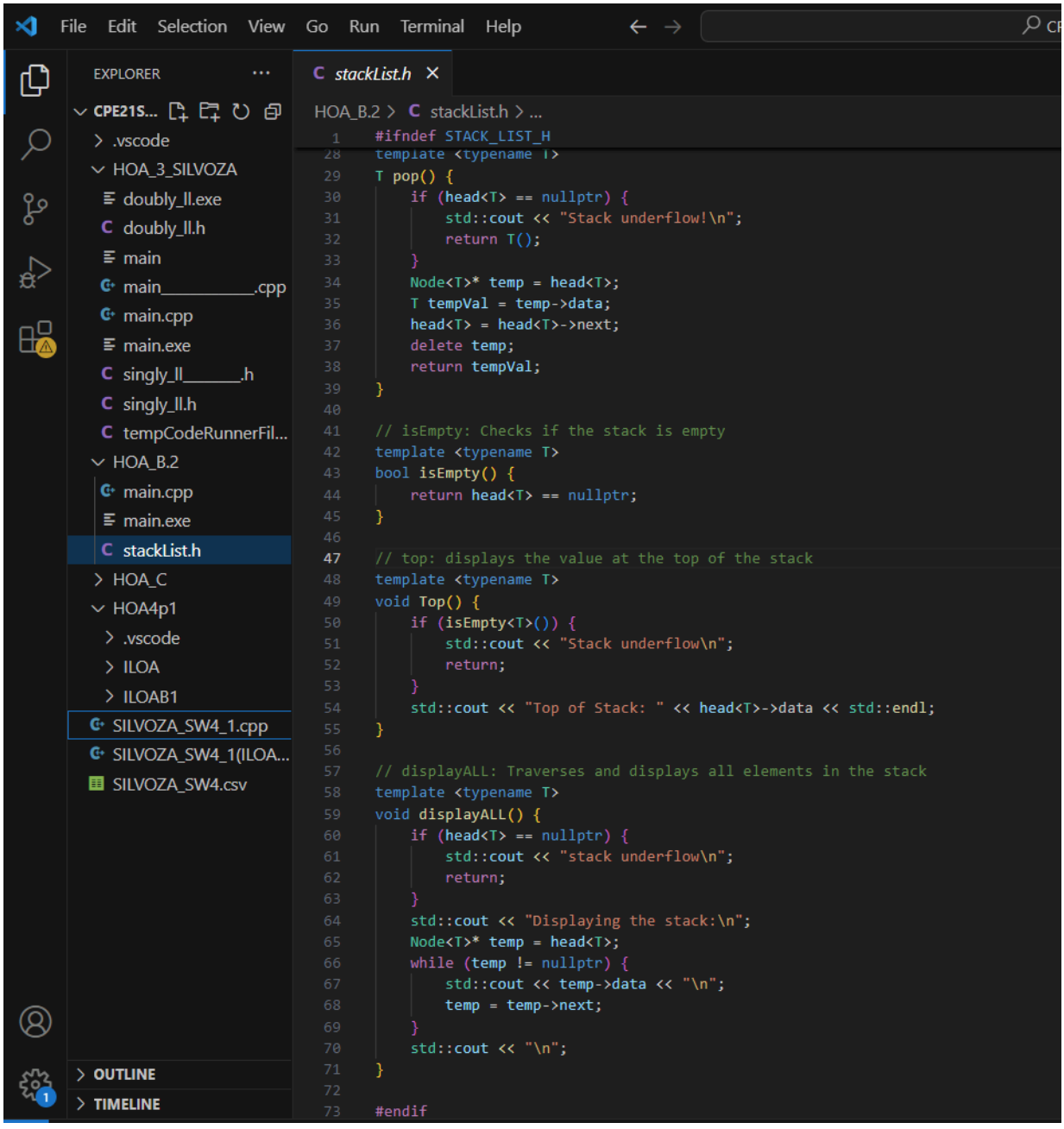
ILOB.2

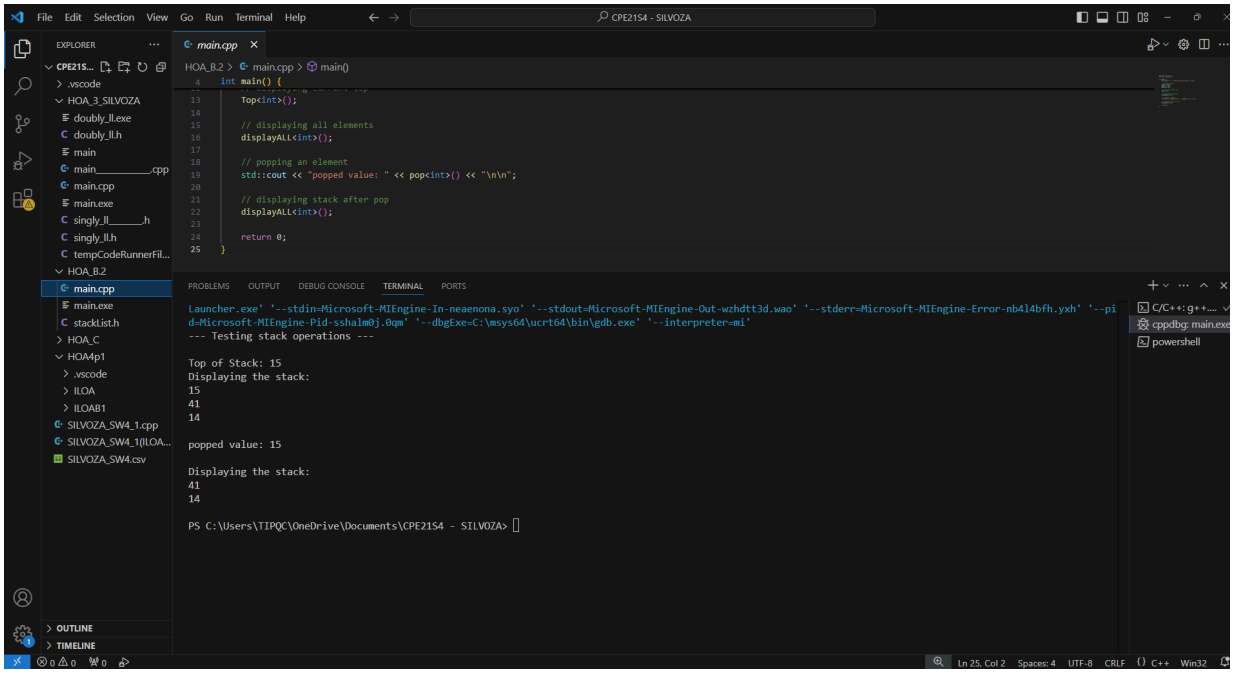



File Edit Selection View Go Run Terminal Help
CPE21S4 - SILVO
EXPLORER
CPE21S...
.vscode
HOA_3_SILVOZA
doubly_ll.exe
doubly_ll.h
main
main_____.cpp
main.cpp
main.exe
singly_ll_____.h
singly_ll.h
tempCodeRunnerFil...
HOA_B.2
main.cpp
main.exe
stackList.h
HOA_C
HOA4p1
.vscode
ILOA
ILOAB1
SILVOZA_SW4_1.cpp
SILVOZA_SW4_1(ILOA...
SILVOZA_SW4.csv
OUTLINE
stackList.h
HOA_B.2 > stackList.h > ...
#ifndef STACK_LIST_H
#define STACK_LIST_H
#include <iostream>
// class node that contains data and pointer to the next node
template <typename T>
class Node {
public:
T data;
Node* next;
};
// pointer for the top of the stack
template <typename T>
Node<T>* head = nullptr;
// push: Adds an element to the top of the stack
template <typename T>
void push(T newData) {
Node<T>* newNode = new Node<T>;
newNode->data = newData;
newNode->next = head<T>;
head<T> = newNode;
}
// pop: removes and returns the top element of the stack
template <typename T>
T pop() {
if (head<T> == nullptr) {
std::cout << "Stack underflow!\n";
return T();
}
Node<T>* temp = head<T>;
T tempVal = temp->data;
head<T> = head<T>->next;
delete temp;
return tempVal;
}
// isEmpty: Checks if the stack is empty
template <typename T>
bool isEmpty() {
return head<T> == nullptr;
}

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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